HMC ONE
AI STRATEGY. BUILT TO SCALE.

CLIENT CASE STUDY · OIL & GAS · SALES & MARKETING

SALES REPORTING · ONE POWER BI MODEL · FISCAL-YEAR FORECAST

Sales Forecasting & BI

Dozens of office sales files, separate masters and competing year definitions became one Power BI model with a fiscal-year calendar, DAX measures and a forecast that sales leaders can read and question.

~~26~~ → **4** hours of monthly reporting effort

48 files
Sales files consolidated per refresh
plus product and sales-group masters

−85 %
Monthly reporting effort
▼ from about 26 hours to 4

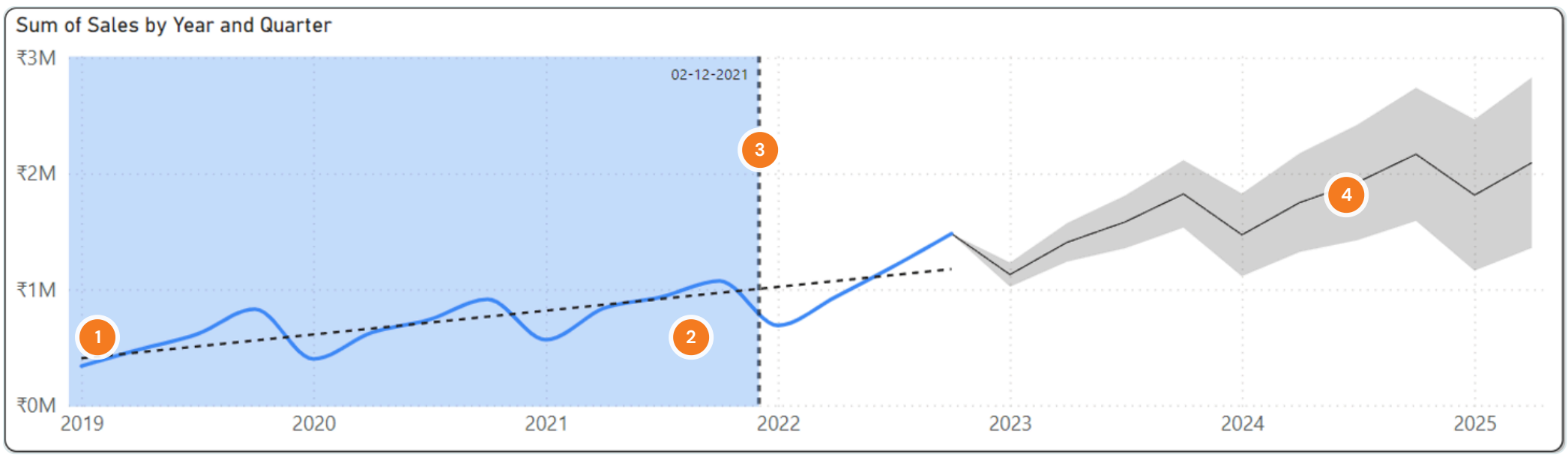
1 fiscal calendar
Year definitions in use
▼ from several spreadsheet versions

12 quarters
Forecast horizon in the dashboard
▲ with confidence bands

Rings: one segment per sales file; 4 of the former 26 hours remain (full ring inside = before); the fiscal year by quarter, April in orange; one segment per forecast quarter. Figures indicative; validated metrics available on request.

01 The forecast page, as delivered

Sum of sales by year and quarter · original Power BI screen, cropped



1 **Actual sales**: about ₹0.4M a quarter in 2019, rising to about ₹1.5M by late 2022

2 **Trend line**: a dashed linear trend drawn through the actual quarters

3 **Cut-off marker** at 02-12-2021; Open Date and Case Origin slicers (cropped out) filter the view

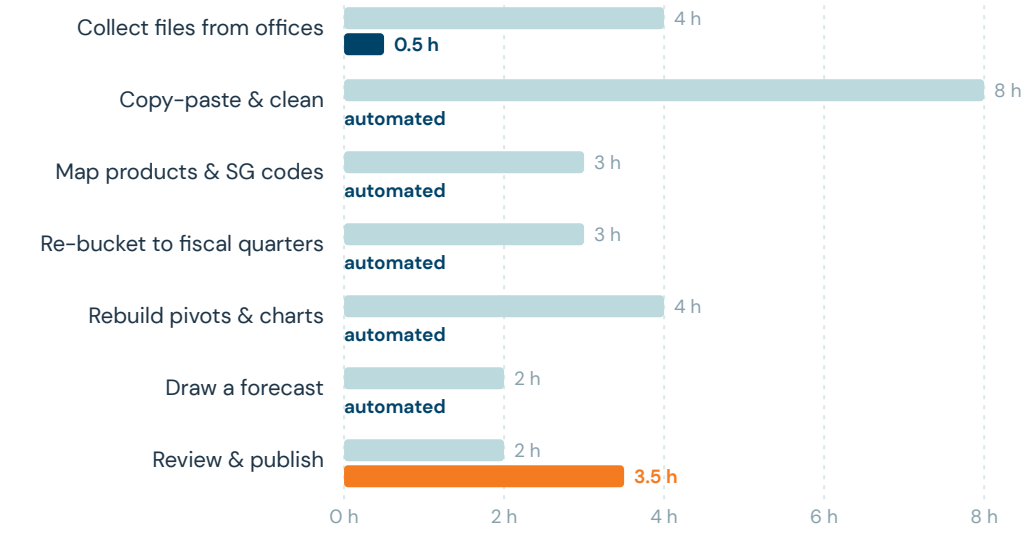
4 **Forecast to 2025**: seasonal, with a grey confidence band that widens with the horizon

02 Days of assembly became a refresh

Impact dashboard · figures indicative

Monthly reporting cycle

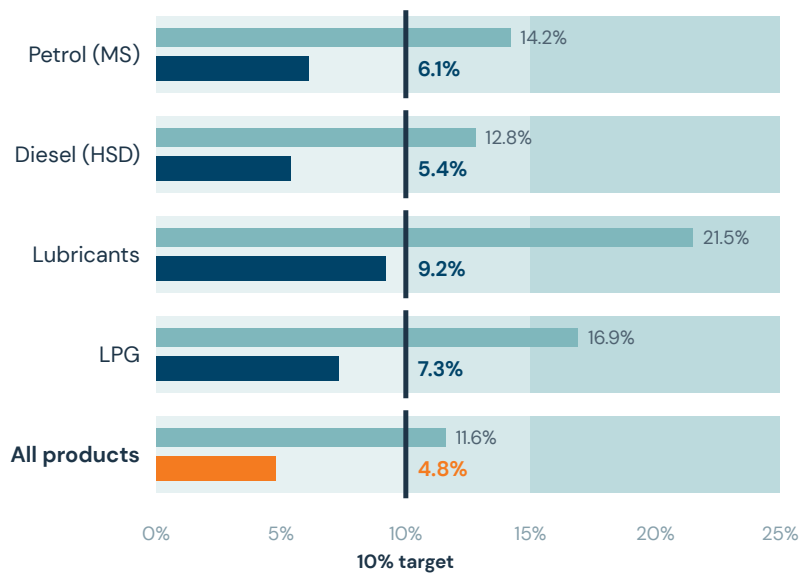
Hours by task, before (light) and after (navy)



26 h → 4 h per monthly reporting cycle
about 3 working days before; refresh and review after

Forecast error by product

Mean absolute % error: spreadsheet trend (light) vs Power BI forecast (navy)

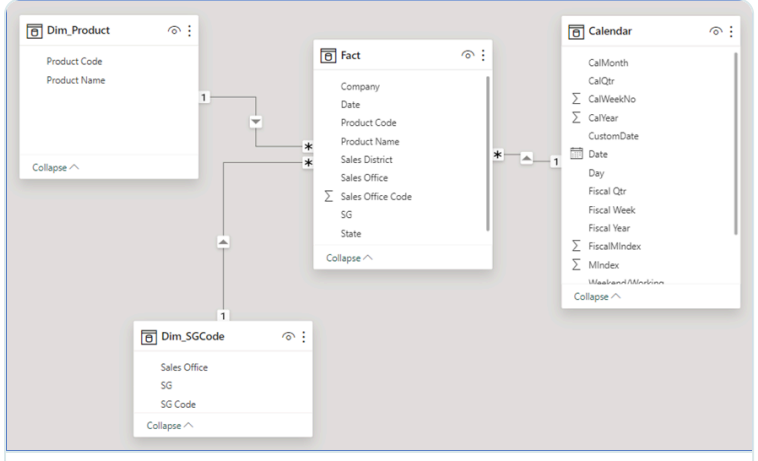


Bands: good (<10%) · fair (10–15%) · poor (15%+)

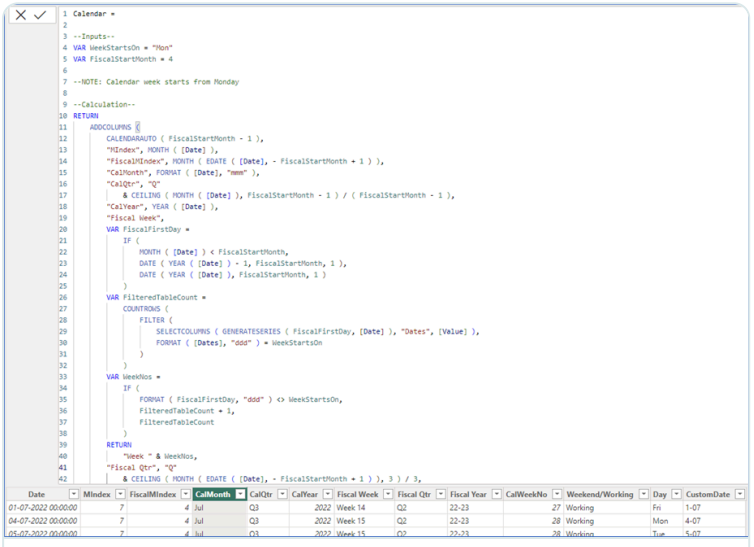
Every product now inside the 10% target. Lubricants, the hardest to call, fell from 21.5% to 9.2%; across all products error dropped from 11.6% to 4.8%.

03 One model under every number

Original screens · data model, calendar DAX and the year it encodes



1 **Star schema**: one Fact table (company, date, product, sales district, office, SG, state) fed by Dim_Product, Dim_SGCode and Calendar, filtering one way



```
Calendar =

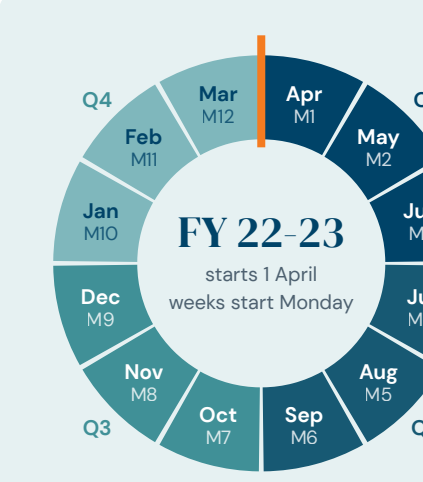
--Inputs--
VAR WeekStartsOn = "Mon"
VAR FiscalStartMonth = 4

--NOTE: Calendar week starts from Monday

--Calculation--
RETURN
    ADDCOLUMNS (
        CALENDARAUTO ( FiscalStartMonth - 1 ),
        "MIndex", MONTH ( [Date] ),
        "FiscalMIndex", MONTH ( EDATE ( [Date], - FiscalStartMonth + 1 ) ),
        "CalMonth", FORMAT ( [Date], "mmm" ),
        "CalQtr", "Q"
            & CEILING ( MONTH ( [Date] ), FiscalStartMonth - 1 ) / ( FiscalStartMonth - 1 ),
        "CalYear", YEAR ( [Date] ),
        "Fiscal Week",
        VAR FiscalFirstDay =
            IF (
                MONTH ( [Date] ) < FiscalStartMonth,
                DATE ( YEAR ( [Date] ) - 1, FiscalStartMonth, 1 ),
                DATE ( YEAR ( [Date] ), FiscalStartMonth, 1 )
            )
        VAR FilteredTableCount =
            COUNTROWS (
                FILTER (
                    SELECTCOLUMNS ( GENERATESERIES ( FiscalFirstDay, [Date] ), "Dates", [Value] ),
                    FORMAT ( [Dates], "ddd" ) = WeekStartsOn
                )
            )
        VAR WeekNos =
            IF (
                FORMAT ( FiscalFirstDay, "ddd" ) <> WeekStartsOn,
                FilteredTableCount + 1,
                FilteredTableCount
            )
        RETURN
            "Week " & WeekNos,
        "Fiscal Qtr", "Q"
            & CEILING ( MONTH ( EDATE ( [Date], - FiscalStartMonth + 1 ) ), 3 ) / 3,
```

Date	MIndex	FiscalMIndex	CalMonth	CalQtr	CalYear	Fiscal Week	Fiscal Qtr	Fiscal Year	CalWeekNo	Weekend/Working	Day	CustomDate
01-07-2022 00:00:00	7	4	Jul	Q3	2022	Week 14	Q2	22-23	27	Working	Fri	1-07
04-07-2022 00:00:00	7	4	Jul	Q3	2022	Week 15	Q2	22-23	28	Working	Mon	4-07
05-07-2022 00:00:00	7	4	Jul	Q3	2022	Week 15	Q2	22-23	28	Working	Tue	5-07

2 **Calendar DAX**: two inputs, WeekStartsOn = "Mon" and FiscalStartMonth = 4, generate fiscal month, week, quarter and year labels



What the calendar fixes. The business year runs April to March, so April is fiscal month 1 and July 2022 lands in Fiscal Q2 of FY 22-23, not calendar Q3.

04 Four pain points, four fixes

Before → after

BEFORE
Scattered data
Product, sales-code and numerous sales files sat in a folder, each in its own shape.

POWER QUERY

AFTER
One folder, appended
Every file read, appended and joined to the masters by Power Query.

BEFORE
No central analytics
Decisions leaned on whichever spreadsheet was closest, or latest.

STAR SCHEMA · DAX

AFTER
One shared model
DAX measures for YoY, FYTD and trend replace static pivots rebuilt monthly.

BEFORE
April, not January
Calendar-year dates and local quarter logic put weeks and quarters in the wrong buckets.

FISCAL CALENDAR

AFTER
One DAX calendar
April start, Monday weeks; parameters documented for future changes.

BEFORE
Manual, every month
About 3 days a month of copying, cleaning and a straight line drawn in Excel.

REFRESH ROUTINE

AFTER
Drop, refresh, review
New files go in the folder; a seasonal forecast with bands updates itself.

05 How HMC ran it

Model once, refresh forever · built with the client's sales analytics team

1 WEEK
1 Discover
Listed every file, master and spreadsheet that fed sales reporting, including several local definitions of quarter and week.

DELIVERABLE
Source inventory and data dictionary

1 WEEK
2 Design
One fact table, three dimensions, one direction of filtering; fiscal year from April, weeks from Monday.

DELIVERABLE
Data model and measure list

3 WEEKS
3 Build
Power Query folder append and master joins, a DAX fiscal calendar, then measures and a forecast with confidence bands.

DELIVERABLE
Power BI model and Forecast dashboard

1 MONTH
4 Adopt
Files drop in, the dashboard refreshes; forecasts reviewed in the monthly business review.

DELIVERABLE
Refresh routine and user walkthrough

ONGOING
5 Scale
New dimensions plug into the same star; calendar reused; forecast extended to product and region level.

DELIVERABLE
Reusable model template

What any sales team can borrow

01
Point at the folder, not the file
A query that reads the whole folder never needs editing when next month's file arrives.

02
Own your calendar
A parameterised fiscal calendar puts every week and quarter in the bucket the business actually uses.

03
Show the band, not just the line
A forecast with its uncertainty, back-tested against real quarters, earns trust that a single line never does.

Built with **Power Query** · **Power BI** (data model, forecasting) · **DAX** | Prepared by Havish M Consulting · Figures indicative; validated metrics available on request.

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